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Department of Computer Applications

FaceTally – Facial Attendance System

MAJOR PROJECT-II
Even Semester
Session 2025-26

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INTRODUCTION

Objective

- Automate student tracking using facial recognition.
- Reduce manual errors and eliminate roll calls.
- Capture, identify, and log attendance instantly.

Problem

- Manual records are time-consuming and prone to proxy.
- Paper logs are hard to analyze and maintain.
- Lack of real-time monitoring and verification.

Solution

Deploy a Python-based system using **FaceNet + SVM** for real-time identification. It integrates with a **MySQL** database to automatically log status, date, and time, ensuring highly accurate and secure records.

How We're Different

- **Contactless:** No physical contact required, addressing hygiene concerns.
- **Anti-Proxy:** Face recognition cannot be faked by sharing cards or IDs.
- **Real-Time Detection:** High-speed using deep learning (FaceNet CNN).
- **Duplicate Prevention:** Blocks re-entry for the same student on the same day.

SDG Alignment

Supports **SDG 4 — Quality Education** by leveraging cutting-edge technology to improve classroom management and institutional efficiency.

Existing Solutions

System	Limitation
Manual Roll Call	Slow, error-prone
Biometric Fingerprint	Contact-based, hygiene
RFID Cards	Proxy-prone, shareable
Basic CCTV	No auto-recognition

TECHNOLOGY USED

Category	Technology
Programming	Python 3.10
GUI Framework	Tkinter (Desktop App)
Computer Vision	OpenCV
Deep Learning	TensorFlow / Keras
ML Classifier	SVM (scikit-learn)
Database	MySQL (XAMPP/MariaDB)

ML Models Used

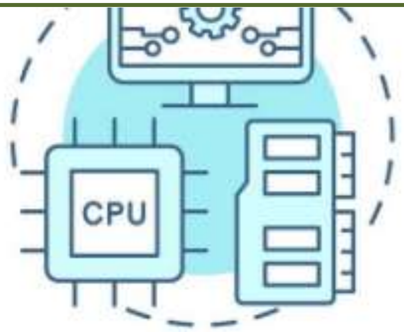
- **FaceNet CNN:** Generates 128-D face embeddings.
- **SVM:** High-performance classifier for recognition.

SYSTEM ARCHITECTURE

Module	Description
Admin Login	Secure username/password authentication
Student Management	Capture face photos & manage records
Feature Extraction	Generate 128-D embeddings via FaceNet
Model Training	Train SVM classifier on extracted data
Take Attendance	Real-time identification & MySQL logging
Attendance Report	View, search, and filter student logs

Database Schema

- **attendance:** ID, Name, Dept, Email, DOJ
- **report:** Log ID, Name, Date, Time, Status
- **login:** Admin credentials



SYSTEM FLOWCHART

1. Camera Input (Webcam captures live video)

2. Image Pre-processing (Grayscale & Haar Cascade Detection)

3. Embedding Extraction (FaceNet generates 128-D vector)

4. SVM Classification (Model predicts identity + confidence)

5. Database Check (Verify daily attendance status)

6. Log Attendance / Alert (Record data or show duplicate warning)

WORKING METHODOLOGY

Step 1: Registration

Admin registers student data and webcam captures **50 photos** per student for accurate training.

Step 2: Extraction

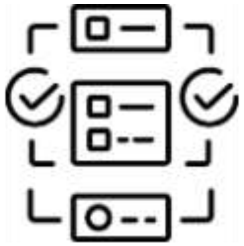
Images are resized to **160x160**. FaceNet processes these to generate unique 128-D vectors saved in a pickle file.

Step 3: Model Training

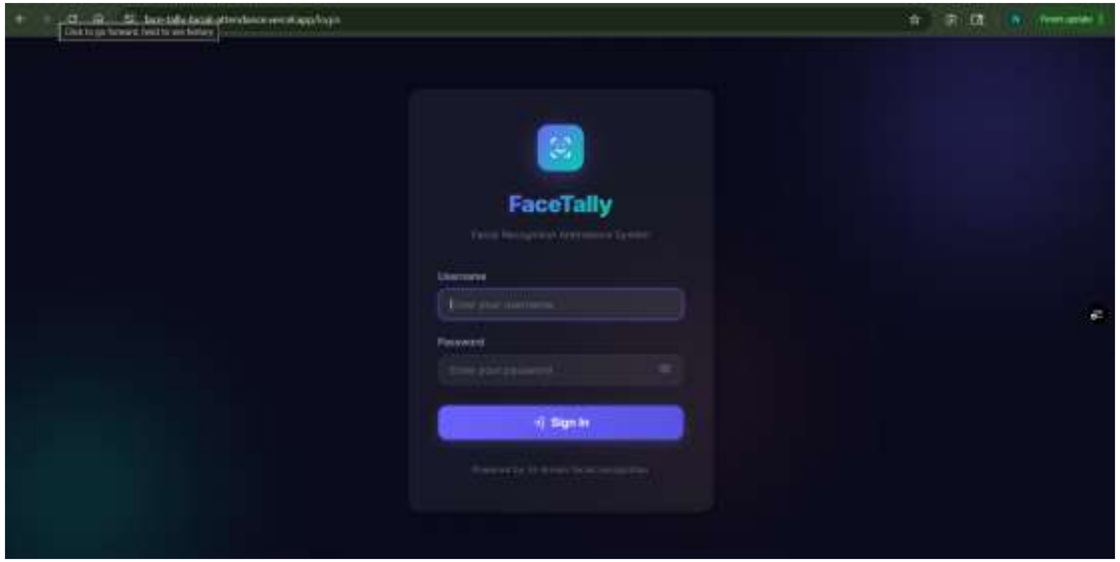
Embeddings are labeled using LabelEncoder. A **Linear SVM** classifier is trained and saved as a recognizer model.

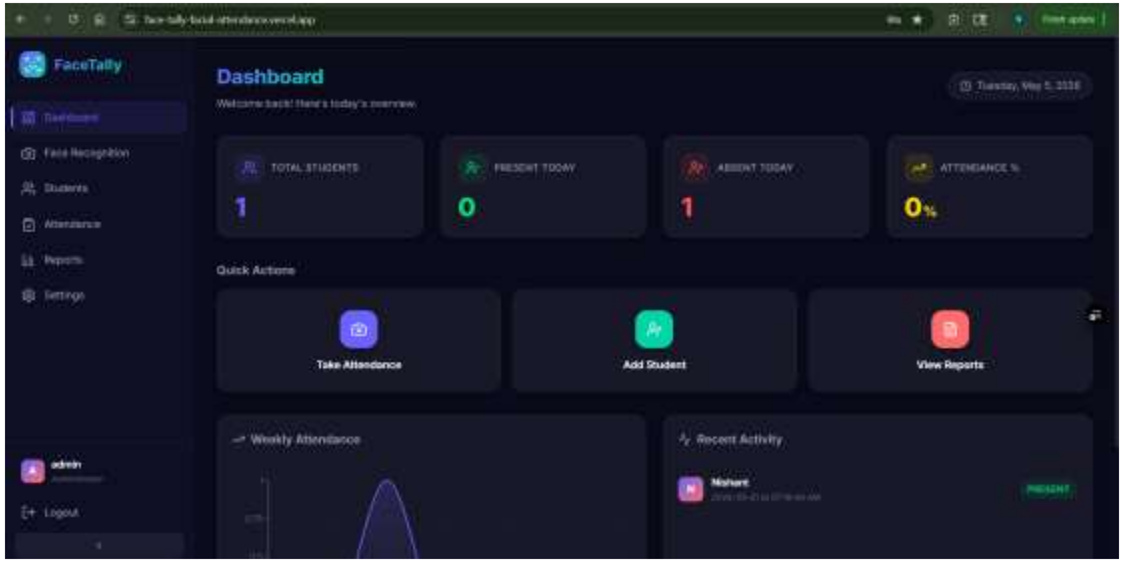
Step 4: Recognition

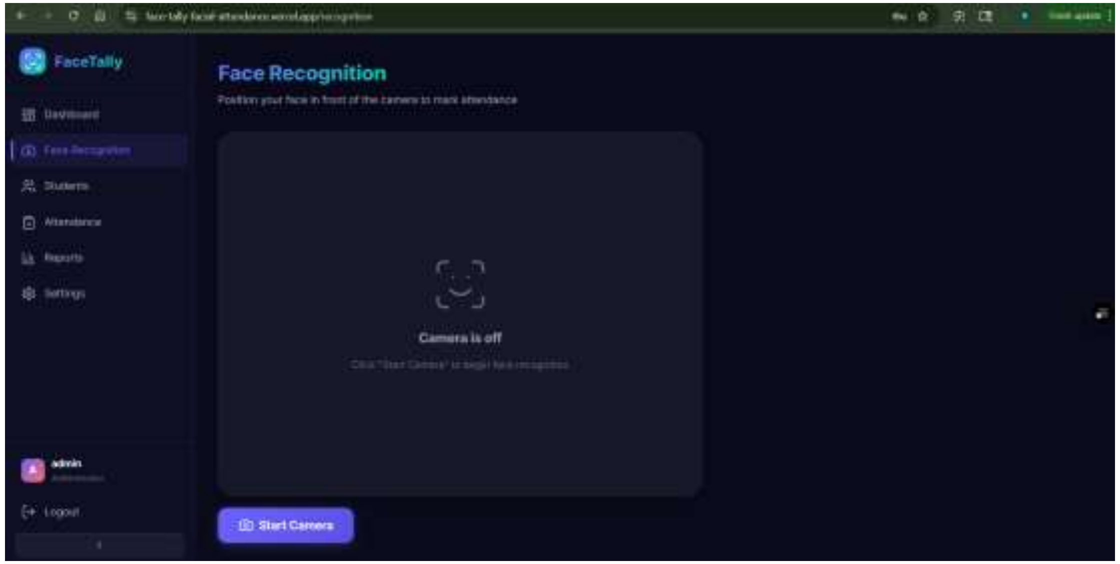
Live feed is processed frame-by-frame. If **Confidence ≥ 60%**, identity is confirmed and logged in MySQL.



SCREENSHOTS







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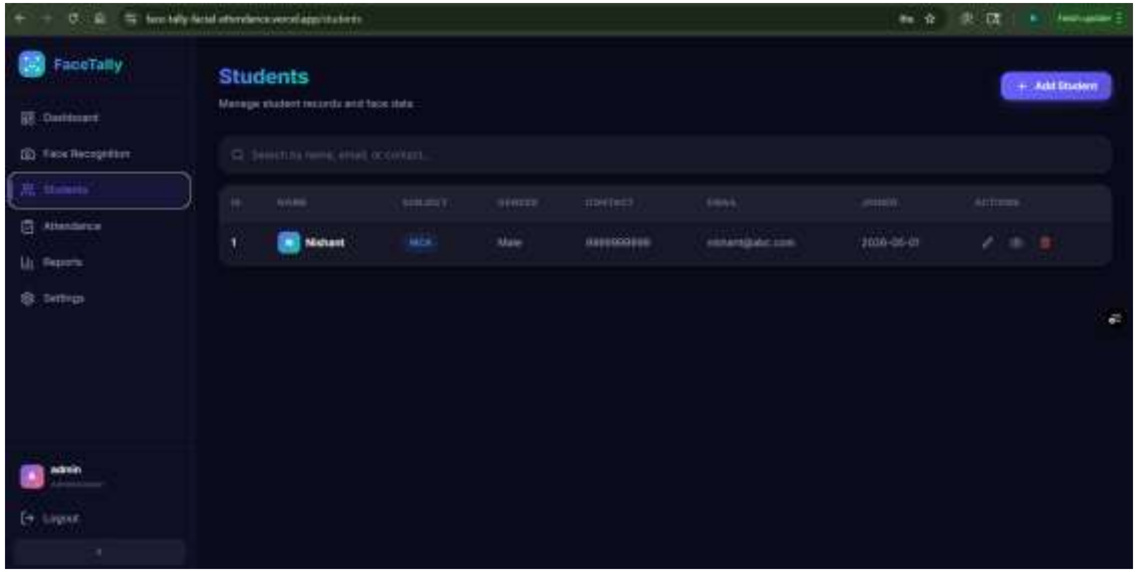
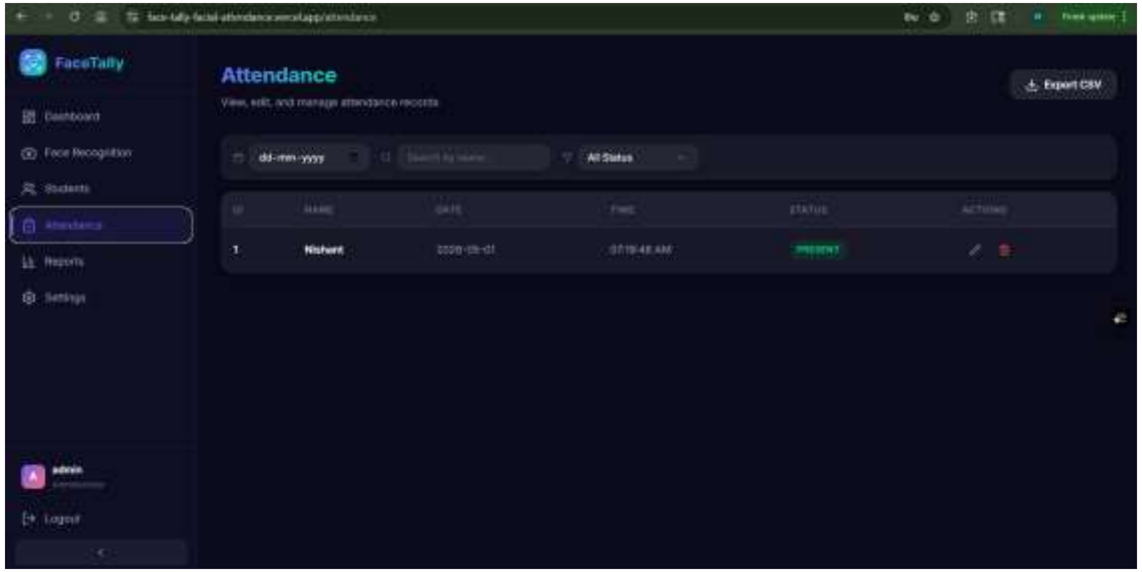
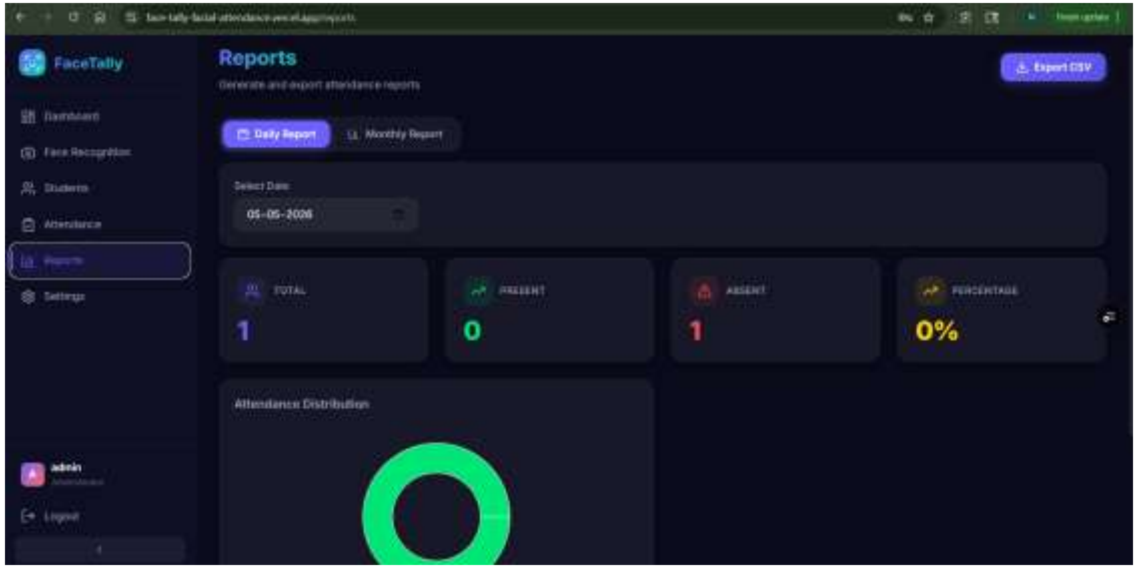
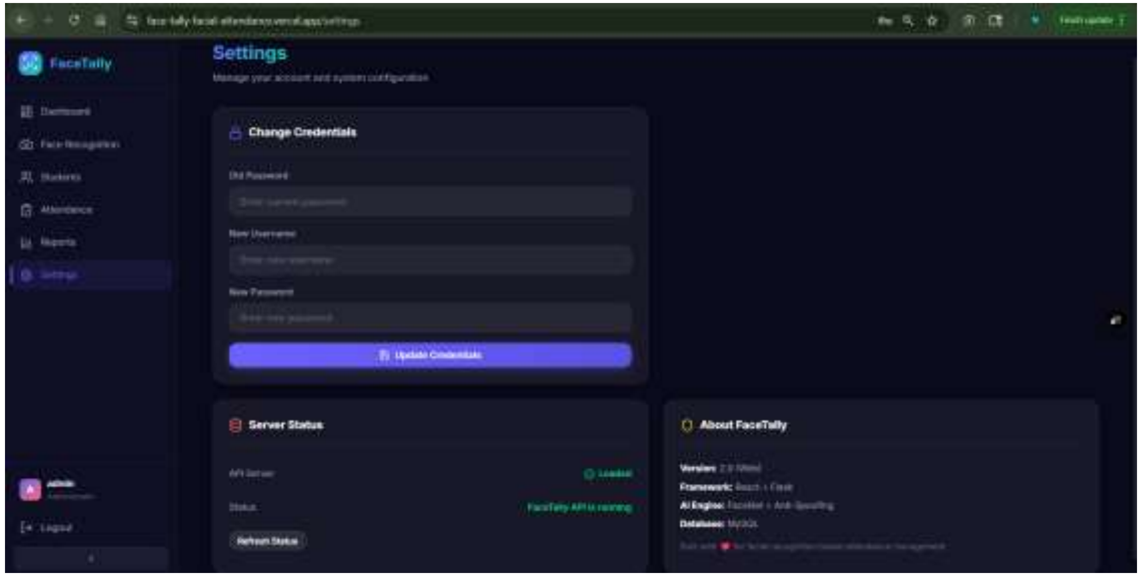


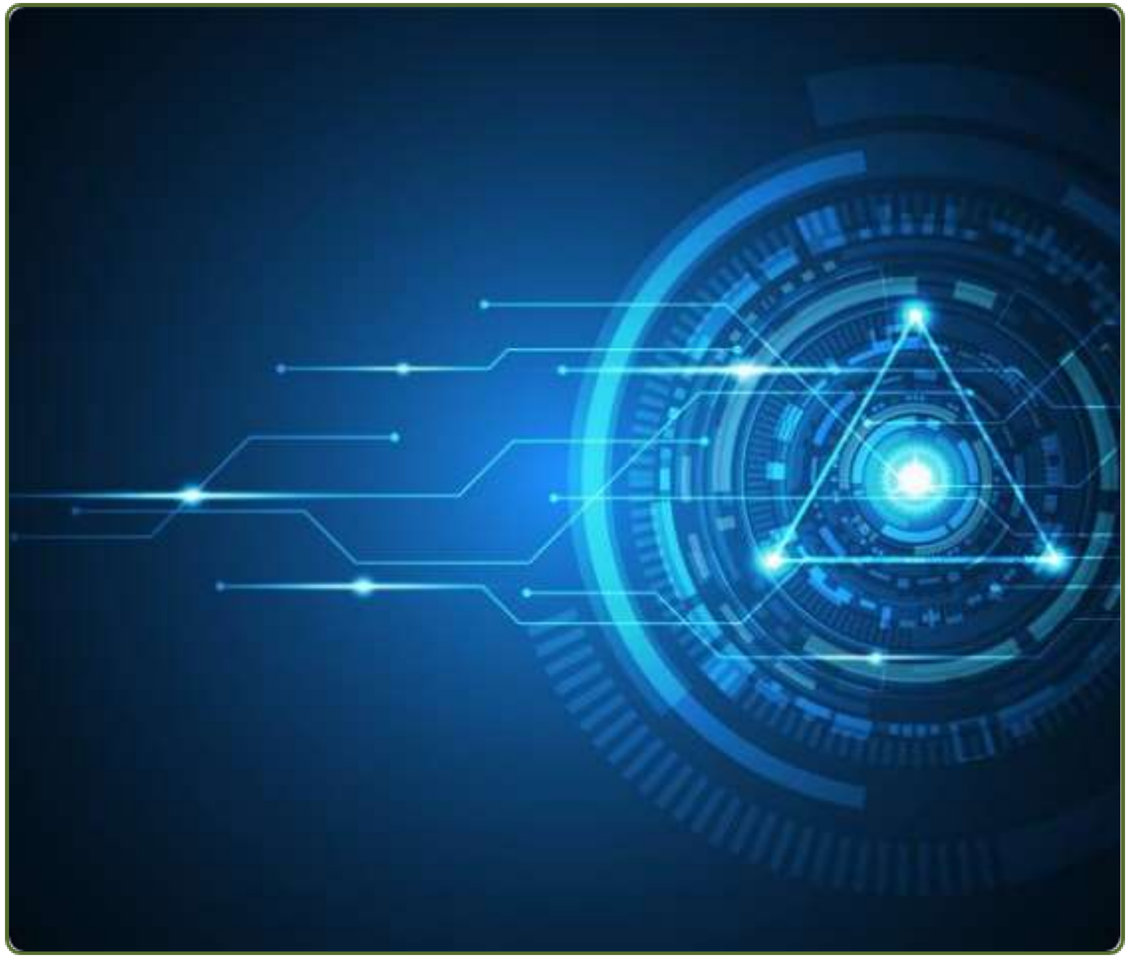






FUTURE SCOPE

- **Anti-Spoofing:** Liveness detection to prevent photo/video-based spoofing.
- **Web Dashboard:** Migrate to Flask/Django for global remote access.
- **Cloud Deployment:** Host database & models for multi-campus usage.
- **Notifications:** Email/SMS alerts via gTTS or SMTP integration.
- **Multi-Camera:** Scale to simultaneous feeds for multiple halls.
- **Analytics:** Graphical heatmaps and attendance trend charts.



CONCLUSION

"The **Face Recognition Based Attendance System** represents a significant advancement in educational infrastructure, ensuring accurate, proxy-free, and efficient tracking using AI and Computer Vision."

Thank You!